

Explainable AI reveals the dominant regulatory factors of Lentinula edodes morphogenesis in factory cultivation

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Abstract: Aiming at the unclear non-linear coupling mechanisms of multi-dimensional "environment-substrate" factors and the lack of precise agronomic trait phenotype prediction models in the factory production of Lentinula edodes under Controlled Environment Agriculture, this study proposes a agronomic trait development prediction framework integrating the Internet of Things (IoT) and eXplainable Artificial Intelligence (XAI). Relying on a greenhouse IoT system, the microclimate data and 11 substrate physicochemical indicators were continuously and dynamically monitored throughout the mushroom growth cycle. After reconstructing the 3D microclimate field via Inverse Distance Weighting (IDW), the predictive performance of 10 machine learning algorithms, including XGBoost, LightGBM, and Multilayer Perceptron (MLP), was systematically evaluated. The results demonstrated that the XGBoost model exhibited optimal generalization ability and robustness in predicting four core phenotypic indicators: cap diameter, cap thickness, stipe diameter, and stipe height, with coefficients of determination R^2 reaching 0.9217, 0.8491, 0.8802, and 0.8524, respectively. To break the "black-box" limitation of high-dimensional ensemble tree models, the SHAP (SHapley Additive exPlanations) framework was introduced for global and local interactive attribution analysis. Quantitative results revealed that substrate parameters such as cellulose content and C/N ratio played an independent, underlying dominant role in determining mushroom phenotypes, while environmental factors like average temperature and CO₂ concentration exhibited a highly coupled non-linear synergistic regulatory mechanism. This study not only achieves high-precision prediction of L. edodes phenotypes under multi-dimensional environmental coupling but also provides a quantitative theoretical basis and an interpretable decision-making tool for green, energy-saving precision environmental control in factory Lentinula edodes farming.

Keywords: Lentinula edodes; XGBoost; SHAP; Machine Learning; Precision Environmental

1. Introduction

As the world's second-largest cultivated edible Lentinula edodes, Lentinula edodes possess both nutritional and medicinal value. Known as the "queen of mushrooms" for their thick flesh, unique flavor, and rich content of polysaccharides and trace elements, they have been reported to exhibit preventive, therapeutic, and immunomodulatory effects (Bisen et al., 2010). The vegetative and reproductive growth of Lentinula edodes is highly susceptible

e to environmental factors, with their quality and yield being the result of interactions between the environment, substrate, and genetics. Key environmental factors such as temperature, humidity, light, CO₂ concentration, and substrate composition play a crucial role in the growth and development of shiitake fruiting bodies. Given its considerable economic significance and well-documented health benefits, *Ledodes* has become a cornerstone species in the global edible mushroom industry, with production systems increasingly transitioning from traditional smallholder farming to large-scale Controlled-Environment Agriculture facilities (Rooyse et al., 2017; Song et al., 2025).

The development of *L. edodes* fruiting body agronomic traits—encompassing cap diameter, cap thickness, stipe diameter, and stipe height—is governed by a complex interplay of genetic determinants, environmental conditions, and substrate physicochemical properties (Jang et al., 2013; Song et al., 2025). Among these factors, ambient temperature, relative humidity, CO₂ concentration, and light intensity have been identified as critical environmental regulators of primordial initiation, fruiting body differentiation, and subsequent agronomic trait expression (Jang et al., 2013). Optimal temperature ranges of 15–25 °C and relative humidity levels exceeding 85% are generally required for normal fruiting body development, while deviations beyond these thresholds can result in marked agronomic trait abnormalities, including cap malformation and stipe elongation defects (Jang et al., 2013; Song et al., 2025). Concurrently, the lignocellulosic substrate matrix—predominantly composed of cellulose, hemicellulose, and lignin—serves as the nutritional foundation for mycelial colonization and reproductive growth, with substrate compositional parameters such as the carbon-to-nitrogen (C/N) ratio and the relative proportions of lignocellulosic constituents directly influencing both yield and agronomic trait quality (Song et al., 2025).

To quantitatively characterize the relationship between environmental drivers and *Ledodes* agronomic trait development and to establish predictive models for phenotypic outcomes, several investigations have been conducted under protected cultivation conditions. Wu et al. (2023) performed a systematic analysis of greenhouse environmental factors in the Loess hilly region of China, identifying the dominant regulatory roles of temperature and humidity in shiitake growth and delineating their optimal ranges for greenhouse production systems. Building upon this qualitative understanding, Li et al. (2021) established dynamic quantitative models for cap diameter expansion and fruiting body fresh weight accumulation as functions of ambient temperature and relative humidity during the fruiting body development stage, thereby advancing the field from empirical description toward mechanistic prediction. More recently, Xia et al. (2024) focused on premium flower mushroom (*Hua-gu*) cultivation, employing grey relational analysis to rank environmental factors by their degree of association with quality-grade formation and subsequently developing a growth model based on accumulated effective temperature that integrated temperature, humidity, CO₂ concentration, and light intensity as coupled input variables. Collectively, these studies have progressively refined the quantitative understanding of environment–growth relationships in *L. edodes* cultivation (Jones et al., 2017; Van Klompenburg et al., 2020). However, existing studies have relied on linear regression models and process-based models to quantify the relationship between environmental factors and shiitake growth, without considering the interactive regulatory effects of environmental factors and substrate composition. In reality, the substrate serves as the foundation for shiitake growth, while environmental factors act as gro

with enablers, forming a complex nonlinear coupling relationship. Current research still faces challenges in addressing nonlinear biological responses and uncertainties in parameterization schemes. The intricate nonlinear relationship between shiitake fruiting body development and environmental variables, as well as substrate composition, significantly compromises model prediction accuracy. Even when studies clarify the impact of temperature and humidity on shiitake fruiting body growth and yield, they often fail to propose quantitative optimization targets for agronomic measures at a localized scale. This limitation poses a challenge for refined production management in shiitake cultivation.

Artificial intelligence (AI), and machine learning (ML) in particular, has emerged as a transformative computational paradigm for integrating and modeling complex, high-dimensional datasets in agricultural systems (Sharma et al., 2020). Advanced ensemble learning algorithms—including XGBoost, LightGBM, CatBoost, and Random Forest—possess the capacity to autonomously discover intricate, nonlinear patterns and high-order feature interactions within large-scale datasets that frequently escape detection by conventional statistical approaches (Van Klompenburg et al., 2020; Sharma et al., 2020). These data-driven methods have demonstrated superior predictive accuracy over traditional linear models across a broad spectrum of agricultural applications, encompassing crop yield forecasting, disease and pest detection, soil property mapping, and greenhouse environmental prediction (Sharma et al., 2020). However, the practical deployment of high-performance ML models in agricultural decision support has been significantly constrained by their inherent "black-box" nature; the opaque internal logic of ensemble and deep learning models undermines user trust and impedes the extraction of actionable, mechanistic insights required for agronomic interpretation and intervention (Arrieta et al., 2020). In response to this interpretability challenge, explainable artificial intelligence (XAI) has rapidly evolved into a prominent subfield of AI research, dedicated to rendering the decision-making processes of complex models transparent, accountable, and comprehensible to domain stakeholders (Lundberg and Lee, 2017; Arrieta et al., 2020). Among the expanding arsenal of post-hoc explanation techniques, SHapley Additive exPlanations (SHAP), grounded in the rigorous axiomatic framework of cooperative game theory, uniquely provides both global feature importance rankings and local, instance-level attribution with formal guarantees of local accuracy, missingness, and consistency (Lundberg and Lee, 2017). When coupled with the TreeExplainer algorithm specifically optimized for tree-based ensemble models, SHAP enables the computationally efficient decomposition of model predictions into exact Shapley values while also quantifying pairwise feature interaction effects (Lundberg et al., 2020). Comparative methodological evaluations have consistently demonstrated that SHAP outperforms alternative explanation methods such as LIME and integrated gradients in terms of explanation stability, local fidelity, and consistency with established domain knowledge (Arrieta et al., 2020; Pai et al., 2025). Consequently, the integration of high-performance ML with SHAP-based interpretability analysis has been successfully operationalized across diverse application domains requiring model transparency, including financial risk assessment, clinical diagnosis, and industrial predictive maintenance. Within the agricultural and plant sciences, recent applications of the ML–SHAP analytical framework have yielded novel mechanistic insights into genotype-to-phenotype prediction in plant breeding programs (Novielli et al., 2024), crop yield determinants under climate variability (Wang et al., 2024; Jagan Mohan et al., 2025), environmental drivers of pes

t and disease outbreak dynamics (Wadhwa and Malik, 2024), and the nonlinear threshold effects of biophysical and management factors on agricultural system performance (Pai et al., 2025).

This study aims to develop and validate a transparent and interpretable artificial intelligence model that integrates substrate composition data and environmental factor data during the growth process of *Lentinula edodes* to predict the growth traits of their fruiting bodies. By employing SHAP interpretability techniques, we seek to elucidate the environmental factors and substrate components that drive the growth of *Lentinula edodes* fruiting bodies.

Additionally, we aim to develop a tool that provides refined management strategies for industrialized shiitake production through real-time predictions and comprehensive visualization of feature contributions.

2. Materials and methods

2.1 Experiment environment and Data Acquisition

The experiment was conducted in Cultivation Chamber No. 6 of a large-scale, intelligent, and standardized *Lentinula edodes* production facility located in Suixian County, Hubei Province, China (113.37°E, 31.73°N). Situated in the transitional zone of the northern subtropical monsoon climate, this location has an average annual temperature of 15.6°C, which ideally satisfies the fundamental thermodynamic requirements for multi-tier shelf cultivation of *L. edodes*. The experimental greenhouse features a semi-circular arched cross-section and is oriented in a standard east-west direction. Its primary structural framework is constructed from high-strength hot-dip galvanized steel pipes, externally clad with a high-transmittance, anti-aging polyethylene (PE) film. The geometric dimensions of the greenhouse specify an 8.0 m span and a 43.0 m depth, with the vault apex reaching a maximum absolute elevation of 3.3 m above ground level. The interior spatial layout utilizes a standardized multi-tier cultivation rack array. The artificial substrate logs are horizontally arranged in a matrix on the shelves, with a uniform vertical spacing of 30 cm maintained between adjacent tiers. To ensure optimal aerodynamic ventilation rates and minimize canopy boundary layer resistance, a 130 cm buffer corridor is allocated between the lateral greenhouse walls and the outer edges of the cultivation racks, while the central longitudinal aisle is maintained at a width of 180 cm.



Fig 1. Experimental Environment for *Lentinula edodes*

2.1.1 Environmental Data Acquisition

To address the inherent spatiotemporal heterogeneity of environmental conditions within the large-scale greenhouse—specifically, localized horizontal cold/hot spots and vertical thermal stratification—a three-dimensional, high-spatiotemporal-resolution environmental sensing

g network was engineered using industrial-grade PRISEN (SN-300YM-WIFI) IoT devices (Ayaz et al., 2019). At the perception layer, 36 high-precision, multi-element meteorological nodes with an IP65 protection rating were orthogonally deployed across the 3D space of the cultivation chamber. The operational ranges and precision thresholds of the sensors were calibrated as follows: air temperature spanning -40 to 80°C (accuracy $\pm 0.5^{\circ}\text{C}$), relative humidity across 0 to 95% RH (accuracy $\pm 0.5\%$), illuminance from 0 to $200,000$ Lux (accuracy $\pm 7\%$), and CO_2 concentration ranging from 0 to 5000 ppm (accuracy $\pm (50 \text{ ppm} + 3\% \text{ F.S.})$). For the data transmission layer, a cluster of 18 wireless data acquisition terminals was established. Each terminal aggregated the baseline data from adjacent sensors via hardwired dual-channel RS-485 buses complying with the ModBus-RTU protocol (Ayaz et al., 2019). Subsequently, these terminals executed high-frequency synchronous sampling at 10-minute intervals and streamed the encrypted data packets in real-time to a cloud server via a 2.4 GHz WiFi network for persistent storage.

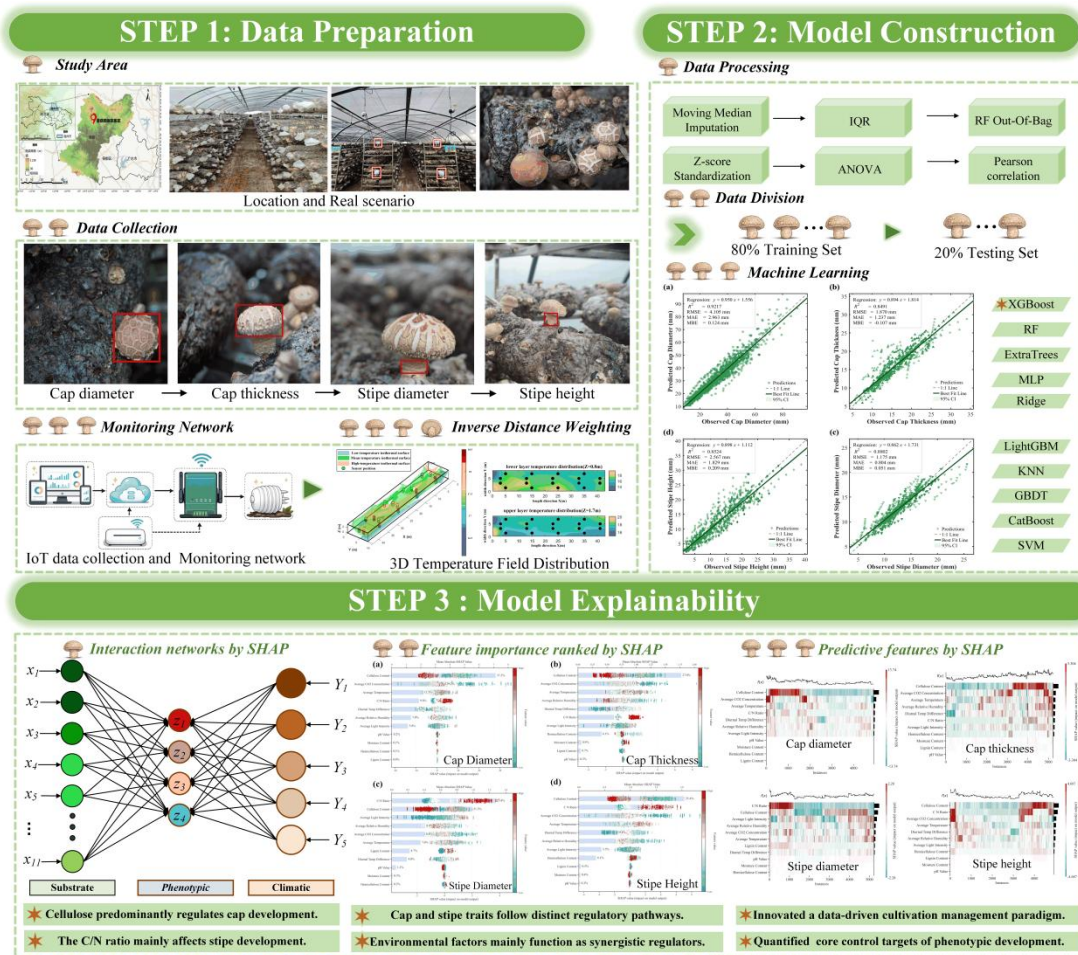


Fig.2 Flowchart of the model framework

2.1.2 Substrate Data Acquisition

Within the vertical projection footprint of the 3D sensory nodes, six representative artificial substrate logs were selected, strictly adhering to a stratified random sampling protocol. During the sampling procedure, 10.0 g substrate core samples were extracted from predetermined locations using sterile instruments. These samples were immediately hermetically sealed in sterile bags and immersed in a dry-ice insulated container for cryogenic quenching to instantly arrest enzymatic

degradation reactions (Gaitán-Hernández et al., 2011). Subsequently, 11 core physicochemical parameters of the substrate matrices were precisely quantified in a biochemical laboratory, encompassing: moisture content, pH, electrical conductivity (EC), carbon-to-nitrogen ratio (C/N), cellulose content, hemicellulose content, lignin content, total nitrogen (TN, g/kg), total carbon (TC, g/kg), potassium (K), and phosphorus (P) (Baktemur et al., 2022).

2.1.3 Phenotypic Data Acquisition

Near the deployed spatial sensory nodes, 36 representative artificial substrate logs (18 in the upper tier and 18 in the lower tier) were selected. Normally developing fruitbodies on these logs were individually tagged and cataloged for in-situ phenotypic measurements. Data acquisition spanned the entire developmental trajectory, commencing at the primordium stage (emergence of the mushroom bud, corresponding to a cap diameter of 1.5–2.0 cm) and concluding upon full fruitbody maturity (Subashini et al., 2024). The monitored agronomic trait metrics encompassed cap diameter, cap thickness, stipe diameter, and overall fruitbody height.

2.2 Data Preprocessing and Statistical Analysis

2.2.1 Data Preprocessing

The research data was acquired from a factory-scale *Lentinula edodes* production facility in Suixian, China, with the production cycle spanning from January 29 to February 15, 2026. A total of 456 mushroom samples were tracked. Each sample underwent continuous monitoring for approximately 18 days from primordium formation to harvest, yielding a cumulative total of 8,208 daily observation records. Each record encompasses environmental monitoring data (temperature, CO₂, humidity, illuminance), substrate assay data (11 chemical components), and agronomic trait measurement data (4 parameters). Raw datasets inherently contain missing and anomalous values, necessitating rigorous preprocessing to enhance model training efficacy. A moving median imputation method was deployed to effectively preserve the underlying trend characteristics of the time-series data. Outliers were identified using the boxplot method, with 1.5 times the Interquartile Range (IQR) established as the threshold (Dazin et al., 2021); data points falling outside the $[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]$ interval were classified as anomalies. To prevent extreme values from skewing the model training, these outliers were replaced with the median of their respective features (Rousseeuw & Hubert, 2011). Post-processing, the proportion of anomalous values was successfully reduced from 2.1% to 0%.

Given the substantial dimensional disparities among the environmental factors and growth status indicators (e.g., CO₂ concentration is measured in ppm, whereas cap diameter is in mm), Z-score standardization was applied to normalize the dataset. This transformation converted all feature values into a standard normal distribution with a mean of 0 and a standard deviation of 1. Finally, a data quality screening protocol based on the Random Forest Out-Of-Bag (OOB) prediction error was executed (Janitza & Hornung, 2018). The 456 sample groups were stratified into three quality tiers: low prediction error samples (OOB score ≤ 0.324), medium prediction error samples (OOB score between 0.324 and 0.746), and high prediction error samples (OOB score ≥ 0.746). This study retained a total of 342 groups (6,156 records) comprising the low and medium error samples to serve as the finalized robust modeling dataset.

2.2.2 Stratified Dataset Partitioning

To rigorously prevent temporal data leakage, the dataset partitioning was strictly executed at the individual mushroom level rather than at the daily record level. Ensuring that the entire 18-day life cycle data of any given mushroom exclusively resides in either the training or testing set. The 342 sanitized modeling sample groups were randomly partitioned into a training set and a testing set at an 8:2 ratio. To guarantee the unbiasedness of the model evaluation and the generalization capability of the subsequent networks, the partitioning process strictly adhered to a stratified sampling strategy based on the sample error distribution profile (Kuhn & Johnson, 2013). This explicitly ensured that the ratio of low-error samples (66 groups) to medium-error samples (208 groups) in the training set—established at 1:3.15—remained highly consistent with the global dataset. Consequently, the training set comprised 4,932 daily records, while the testing set contained 1,224 daily records.

Subsequently, Z-score standardization (StandardScaler) was applied to map the data feature matrix. Assuming the input feature matrix is $\mathbf{X} \in \mathbb{R}^{n \times p}$ (where n is the sample size and p is the feature dimension), the standardization formula is mathematically defined as:

$$x_{ij}^* = \frac{x_{ij} - \mu_j}{\sigma_j} \quad (1)$$

where μ_j and σ_j represent the mean and standard deviation of the j -th feature computed exclusively within the training set, respectively. To rigorously prevent data leakage that could artificially inflate predictive accuracy, the standardization parameters μ_j, σ_j were strictly fitted only on the training set and were subsequently applied to transform the testing set using identical mapping criteria.

2.2.3 Statistical Analysis

Analysis of Variance (ANOVA) revealed that all 20 candidate variables, alongside the target variables, exhibited highly significant differences $P < 0.001$ across distinct fruitbody developmental stages. This substantiates that the selected agronomic trait indicators, environmental parameters, and substrate variables robustly characterize the physiological evolution inherent to varying developmental phases. Furthermore, Pearson correlation analysis identified severe multicollinearity among specific substrate features (Van Klompenburg et al., 2020).

Table 1 Statistical significance analysis of various experimental variables

Variable Category	Variable Name & Symbol (Unit)	P-value	Significance
Agronomic Trait	Cap diameter, D_{cap}/mm	1.3919e-215	***
	Stipe height, H_{sti}/mm	1.5342e-159	***
	Cap thickness T_{cap}/mm	5.0059e-101	***
	Stipe diameter D_{sti}/mm	2.8098e-59	***
Environmental Parameters	Diurnal temperature difference, $\Delta T/^{\circ}\text{C}$	5.6984e-39	***
	Average relative humidity, $\text{RH}/\%$	4.5442e-37	***
	Average temperature, $T_{avg}/^{\circ}\text{C}$	2.0983e-33	***
	Average CO_2 , $C_{\text{CO}_2}/\text{ppm}$	7.6405e-25	***
	Average illuminance, L/Lux	1.0661e-08	***

Substrate Parameters	Acidity/Alkalinity, pH	1.0207e-28	***
	Total carbon, TC	2.9979e-28	***
	Carbon content, C/%	3.2891e-28	***
	Carbon-to-nitrogen ratio, C/N	2.0503e-51	***
	Nitrogen content, N/%	3.4139e-49	***
	Hemicellulose, $w_{\text{hem}}/(\text{mg}\cdot\text{g}^{-1})$	1.4451e-18	***
	Cellulose, $w_{\text{cel}}/(\text{mg}\cdot\text{g}^{-1})$	2.9094e-69	***
	Lignin, $w_{\text{lig}}/(\text{mg}\cdot\text{g}^{-1})$	3.0407e-14	***
	Potassium, K	6.5516e-10	***
	Total nitrogen, TN	7.1765e-06	***
	Moisture content, $w_{\text{H}_2\text{O}}/\%$	4.9629e-05	***

Note: *** indicates $P < 0.001$. Analysis of variance (ANOVA) revealed that all 20 candidate variables exhibited highly significant differences across varying fruitbody developmental stages.

For instance, carbon content (C%) and total carbon demonstrated a perfect positive correlation; the C/N ratio and nitrogen content (N%) exhibited a highly significant negative correlation; and pH values were strongly positively correlated with potassium content. To eradicate underlying data redundancy, mitigate collinearity-induced interference during model training, and strictly prevent interpretability distortions in downstream SHAP attributions, overlapping redundant variables—namely carbon content, nitrogen content, total carbon, total nitrogen, and potassium—were decisively excluded. Ultimately, the feature engineering process screened and retained 5 environmental factors (average temperature, diurnal temperature difference, average illuminance, average relative humidity, and average CO_2 concentration) alongside 6 substrate physicochemical factors (C/N ratio, lignin, cellulose, hemicellulose, pH, and moisture content) to serve as the core predictive driving variables.

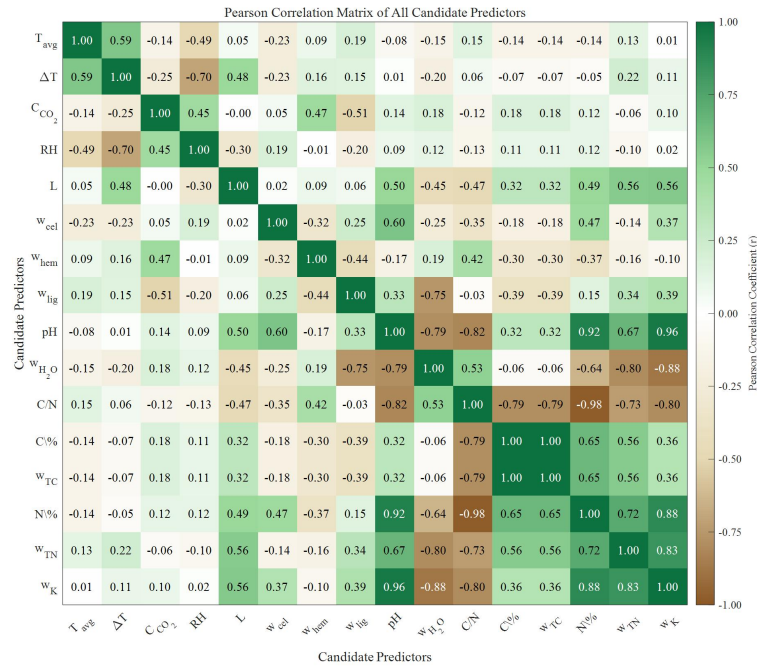


Fig. 3 Pearson correlation matrix of all candidate predictors

2.3 Spatial Reconstruction of Environmental Factors

Within the large-scale cultivation greenhouse, environmental factors such as temperature, humidity, and CO₂ concentration exhibit pronounced spatial heterogeneity and vertical thermodynamic stratification due to structural shading, crop canopy obstruction, and ventilation non-uniformity (Bojacá et al., 2009). Consequently, conventional global averaging methods inherently fail to capture the authentic environmental conditions at the specific locations of individual mushroom samples, potentially introducing systematic bias into downstream modeling. To accurately estimate the environmental parameters at the exact physical coordinates of each monitored mushroom individual, this study introduces the Inverse Distance Weighting (IDW) spatial interpolation algorithm (Ferentinos et al., 2017). Through spatial matrix mapping operations, an environmental feature estimation framework is established, bridging the gap from discrete wireless sensor network observations to continuous environmental variables at target biological individuals. The IDW algorithm was selected for generating the individual-level environmental input features due to its strictly localized distance-decay assumption—which aligns with the physical expectation that spatially proximate sensors better represent the environmental conditions at a target sample location—and its higher computational efficiency for real-time IoT node mapping.

Given the significant environmental gradient variations along the vertical axis within the cultivation chamber, physical topological constraints were initially imposed to strictly circumvent spatial estimation distortions caused by naive cross-layer interpolation. The preprocessed 36 spatially discrete sensor nodes were rigorously partitioned into an upper-layer sensor subset (S_{upper}) and a lower-layer sensor subset S_{lower} . For any target mushroom individual m fixed at the absolute 3D coordinate $P_m(x_m, y_m, z_m)$, only active co-layer sensor nodes (with the number of single-layer nodes $N \leq 18$, after excluding anomalous or inactive reference nodes) were incorporated into the interpolation benchmark set. Assuming the 2D projected coordinate of the i -th co-layer sensor is $P_i(x_i, y_i)$, the horizontal Euclidean distance $d_{i,m}$ between mushroom individual m and sensor i is defined as:

$$d_{i,m} = \sqrt{(x_m - x_i)^2 + (y_m - y_i)^2} \quad (2)$$

The core physical assumption of the IDW algorithm adheres to Tobler's First Law of Geography, which posits that spatially proximate geographical entities share stronger attribute correlations (Tobler, 1970). The environmental contribution weight of a sensor to the target mushroom is inversely proportional to the p -th power of their spatial distance. The distance-decay-based weight formulation is defined as:

$$w_{i,m} = \frac{d_{i,m}^{-p}}{\sum_{j=1}^N d_{j,m}^{-p}} \quad (3)$$

where $w_{i,m}$ denotes the microenvironmental impact weight of the i -th sensor on mushroom m , and p is the power parameter controlling the distance decay rate. Calibrated by empirical greenhouse aerodynamic smoothness parameters, this study set $p = 2$. Consequently, the spatial weight row vector for the target individual m can be formulated as $W_m = [w_{1,m}, w_{2,m}, \dots, w_{N,m}]$, satisfying the condition $\sum_{i=1}^N w_{i,m} = 1$.

Subsequently, time-aligned environmental observation data were extracted. Let E_{temp} , E_{CO_2} , and E_{hum} represent the observation column vectors of average temperature, average CO₂

concentration, and average humidity, respectively, aggregated by the N co-layer sensors within a given measurement time window. Taking the estimation of average temperature as an example, the environmental observation column vector $E_{\text{temp}} \in \mathbb{R}^{N \times 1}$ is defined as:

$$E_{\text{temp}} = [T_1, T_2, \dots, T_N]^T \quad (4)$$

Ultimately, the individualized average temperature estimate T_m at the specific coordinates of mushroom m can be precisely calculated via the dot product of the spatial weight vector and the environmental feature observation vector:

$$T_m = W_m \cdot E_{\text{temp}} = \sum_{i=1}^N w_{i,m} T_i \quad (5)$$

Similarly, the individual-specific average CO₂ concentration and average humidity can be accurately resolved by substituting the corresponding observation column vectors. Through the aforementioned spatial weight matrix mapping strictly subjected to topological constraints, this 3D reconstruction process maximally preserves authentic in-situ local microhabitat characteristics. This provides the downstream machine learning predictive networks with foundational physical input features possessing both high spatial resolution and robust biological interpretability.

2.3.1 Environmental Spatial Variability

To validate the spatial heterogeneity of environmental conditions within the greenhouse and provide a physical basis for the IDW-based individual environmental mapping approach, this study analyzed the spatial distribution patterns of key environmental factors across the 36-sensor monitoring network. Quantitative analysis of the multi-point measurements confirmed the presence of pronounced horizontal heterogeneity and vertical thermodynamic stratification within the facility greenhouse (Bojacá et al., 2009).

From a whole-greenhouse perspective, the mean temperature of the upper cultivation tier ($L_2, z = 1.7m$) was 16.52°C, showing only a small difference from the 16.31°C recorded in the lower tier ($L_1, z = 0.8m$). However, this small difference between the overall means concealed substantial variability at finer spatial and temporal scales. During periods of intense solar radiation, solar heat accumulation and buoyancy-driven convection caused temperatures in some upper-tier regions to exceed those in the corresponding lower tier by nearly 2°C. Conversely, nocturnal long-wave radiative heat loss through the greenhouse roof occasionally produced a temperature inversion, during which the lower tier became warmer than the upper tier. These dynamic vertical reversals demonstrate the limited spatiotemporal representativeness of conventional environmental characterization based on a single fixed sensor or a whole-greenhouse average.

As illustrated in Fig. 4, the temperature field exhibited pronounced horizontal heterogeneity. A longitudinal temperature gradient of approximately 2°C was observed from the greenhouse entrance toward the distal end, potentially resulting from the attenuation of ventilation airflow and spatial differences in solar heat input. The central cultivation racks were approximately 0.5°C warmer than the peripheral racks, indicating a boundary heat-loss effect associated with the greenhouse sidewalls. In addition, a localized high-temperature region, with a peak temperature approaching 20.7°C, occurred in the upper tier of the middle-to-rear section of the greenhouse. This region may have resulted from restricted airflow, prolonged solar heat input, and localized sensible heat accumulation (Bojacá et al., 2009a).

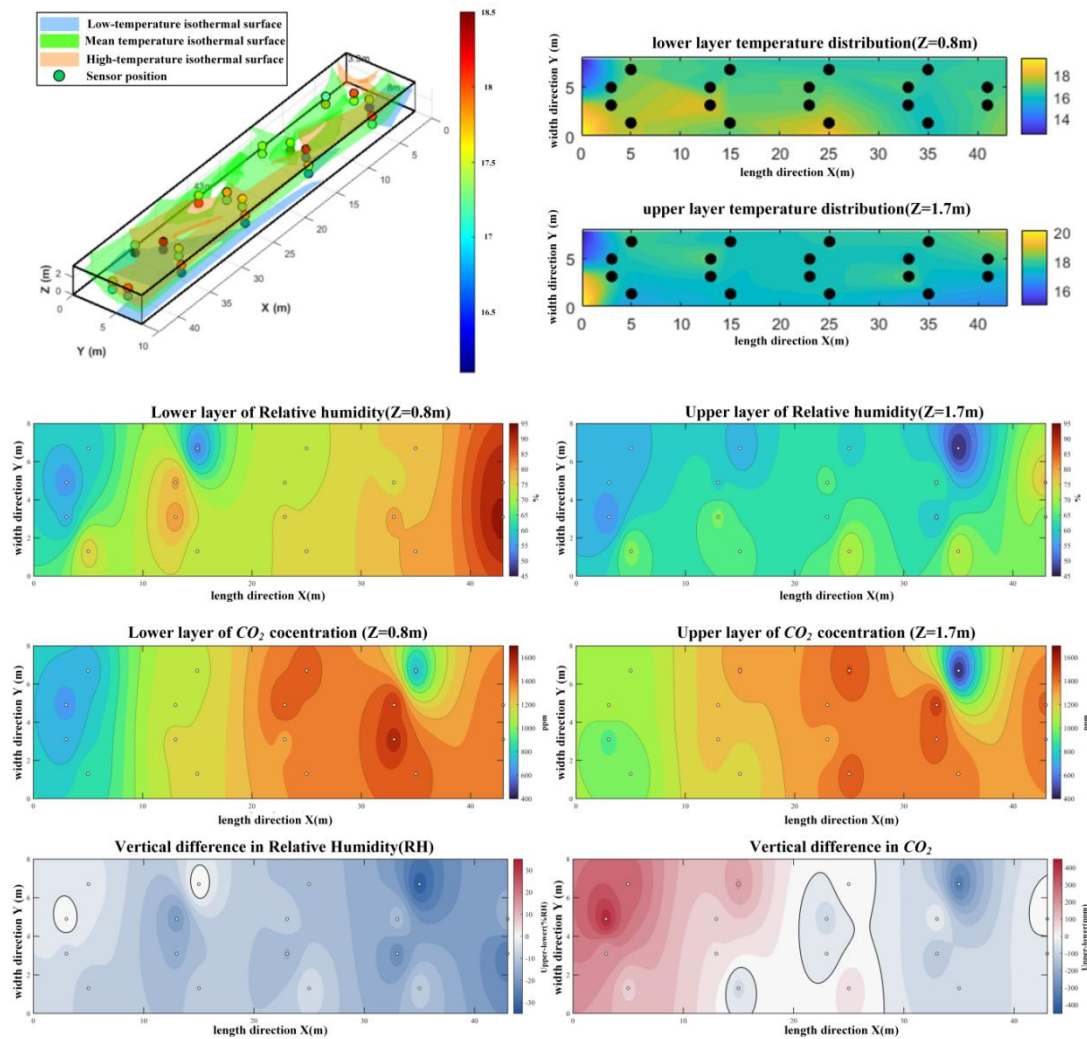


Fig. 4 3D Temperature Field Distribution Diagram

The temperature-field analysis demonstrates that the environmental conditions experienced by individual shiitake mushrooms (*Lentinula edodes*) depend strongly on their spatial positions. Failure to account for such microclimatic variability may reduce the representativeness of environmental data assigned to individual organisms and may consequently bias the analysis of environment–growth relationships (Bojacá et al., 2009b). Temperature, relative humidity, and CO₂ concentration can be treated as spatially continuous scalar variables because their distributions are governed by continuous transport processes, including heat transfer, ventilation airflow, water-vapor transport, transpiration, gas diffusion, and biological respiration. Therefore, inverse distance weighting (IDW) was applied to these three variables to estimate the environmental conditions at each individual *L. edodes* location

Under the IDW framework, measurements from multiple surrounding sensors were integrated according to their distances from the target *L. edodes* individual, with closer sensors assigned greater weights. This distance-decay principle is consistent with the fundamental formulation of IDW and the assumption that observations located closer together generally exhibit greater spatial similarity (Shepard, 1968; Tobler, 1970). Compared with directly assigning a whole-greenhouse average or the measurement from a single fixed sensor to all individuals, this approach provides a more spatially representative estimate of the local growth environment.

Light intensity was processed differently because its spatial distribution within the multi-tier cultivation system does not satisfy the spatial-continuity assumption underlying IDW. Unlike temperature, relative humidity, and CO₂ concentration, which are transported and redistributed through airflow, diffusion, and convection, local irradiance is strongly controlled by greenhouse covering materials, rack geometry, cultivation bags, structural shading, direct-beam obstruction, and the relative positions of light sources. Previous studies have demonstrated that greenhouse structures and shading elements can substantially alter both the intensity and spatial distribution of radiation within greenhouses (Kittas et al., 1999; Teitel et al., 2012).

Applying IDW to light intensity would artificially smooth these discontinuities and could assign contributions from sensors located under substantially different shading conditions, producing estimates that do not reflect the actual light environment experienced by an individual mushroom. Therefore, the light intensity of each mushroom was represented by the measurement from its nearest corresponding light sensor. Because this sensor was located close to the mushroom and shared a similar rack position and shading environment, its measurement was considered a reasonable proxy for the mushroom's actual local irradiance.

Accordingly, the proposed environmental data-processing framework used layer-constrained IDW to reconstruct the spatially continuous fields of temperature, relative humidity, and CO₂ concentration, while using the nearest spatial sensor to characterize the locally discontinuous light environment. This variable-specific strategy improved the spatial representativeness of the environmental data assigned to individual mushrooms and ensured that each processing method was consistent with the physical characteristics of the corresponding environmental factor.

2.4 Model Evaluation Metrics

To objectively quantify the predictive accuracy of various machine learning algorithms and rigorously screen for the optimal benchmark predictive engine, this study introduced four core statistical evaluation metrics: Coefficient of Determination (R^2), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Bias Error (MBE). (Botchkarev, 2019). The specific mathematical formulations for these metrics are defined as follows:

$$R^2 = 1 - \frac{\sum_{j=1}^4 \sum_{i=1}^n (y_{ij} - \hat{y}_{ij})^2}{\sum_{j=1}^4 \sum_{i=1}^n (y_{ij} - \bar{y}_j)^2} \quad (6)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (7)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (8)$$

$$MBE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \quad (9)$$

where y_i represents the actual observed value of the target variable; $\hat{y}_i$ denotes the model's predicted value; $\bar{y}$ is the sample mean of the actual observations; and n indicates the total number of experimental samples. For the four-target prediction tasks concerning mushroom agronomic traits ($j = 1, 2, 3, 4$), the overall global R^2 amalgamates the sum of squared prediction residuals across all targets, whereas the independent R^2 and RMSE for each specific target are

calculated synchronously and separately.

The statistical significance of each metric is detailed as follows: R^2 gauges the model's explanatory capacity for the variance in target variables (i.e., goodness of fit), with values closer to 1 indicating superior fitting precision. MAE evaluates the average absolute magnitude of the errors between predicted and actual values. MBE quantifies the long-term systematic directional bias of the predictive network (i.e., overall overestimation or underestimation), with absolute values closer to zero being highly desirable. RMSE measures the sample standard deviation of prediction errors; due to its squared penalty mechanism, it is inherently more sensitive to localized, abnormally large extreme errors, whereby a smaller RMSE value signifies stronger global anti-noise robustness of the model.

2.5 Construction of the Machine Learning

To address the high dimensionality, profound non-linearity, and spatiotemporal heterogeneity inherently prevalent in agricultural IoT sensory data, this study constructed a candidate benchmark model library encompassing 10 core machine learning architectures (Sharma et al., 2020). Through multi-dimensional broad-spectrum cross-evaluation, the objective was to rigorously screen for the underlying predictive engine with the optimal anti-noise robustness and generalization capability, thereby providing a high-fidelity mathematical mapping foundation for downstream SHAP-based physical mechanism explanations. The 10 selected representative algorithms are categorized into the following three major frameworks:

(1) Tree-Based Ensemble Learning Frameworks: Given their inherent superiority in processing multi-dimensional tabular agricultural data and capturing complex feature interactions, two primary branches were integrated. The **Boosting family**, encompassing XGBoost, (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), GBDT (Friedman, 2001), and CatBoost (Prokhorenkova et al., 2018), employs a sequential iterative strategy to continuously fit residuals utilizing the negative gradient of the loss function, significantly elevating the predictive upper bound under high-dimensional environmental combinations. The **Bagging family**, comprising Random Forest (RF) (Breiman, 2001) and ExtraTrees (Geurts et al., 2006), substantially enhances the model's tolerance to greenhouse microclimatic sensor noise and fortifies anti-overfitting capacities by building multiple mutually independent decision trees and averaging their ensemble outputs.

(2) Classical Statistical and Non-linear Boundary Algorithms: To comparatively evaluate the linear and non-linear spatial distribution characteristics of the mushroom ecological data, specific baseline models were established. **Support Vector Machine (SVM)** (Cortes & Vapnik, 1995), utilizing the Radial Basis Function (RBF) kernel, aims to delineate the optimal hyperplane within a high-dimensional feature space to resolve complex non-linear boundaries between environments and substrates. **K-Nearest Neighbors (KNN)** (Cover & Hart, 1967), representing non-parametric heuristic learning, executes state predictions relying on distance metrics of environmental samples within the feature space. **Ridge Regression** (Hoerl & Kennard, 1970), serving as a linear benchmark incorporated with an L_2 regularization penalty, is primarily utilized to reversely assess whether severe multicollinearity perturbations remain within the feature variable library.

2.6 SHAP

The SHAP method decomposes the model's predicted value $f(x_i)$ for sample i into the sum of the Shapley values of each feature (Lundberg & Lee, 2017):

$$f(x_i) = \phi_0 + \sum_{j=1}^p \phi_j(x_i) \quad (10)$$

where ϕ_0 is the baseline predicted value (the mean of the predicted values of all samples); $\phi_j(x_i)$ is the Shapley value of feature j for sample i , representing the marginal contribution of this feature relative to the average prediction.

In this study, TreeExplainer (with the feature perturbation strategy set to Interventional) was employed to calculate the SHAP values, using all samples from the testing set as the background dataset (Lundberg et al., 2020). The global importance of a feature is defined as the mean of the absolute SHAP values across all testing samples:

$$I_j = \frac{1}{n} \sum_{i=1}^n |\phi_j(x_i)| \quad (11)$$

The Interventional strategy breaks the conditional dependencies among features, thereby avoiding the distortion of SHAP value allocation caused by collinear features.

3. Experimental results and analysis

3.1 Result of machine study

Table 2 Comparison of hyperparameters and evaluation metrics among different predictive models

Model	Parameters	Overall R ²	R ² D_{cap}	R ² T_{cap}	R ² D_{sti}	R ² H_{sti}	RMS E/mm	MAE/ mm	MBE/m
LightGBM	n=300,d=10,η=0.05	0.9039	0.9235	0.8441	0.8749	0.8483	2.66	1.76	0.0991
XGBoost	n=300,d=7,η=0.098	0.9038	0.9217	0.8491	0.8802	0.8524	2.66	1.71	0.0691
KNN	K=10,dist	0.8995	0.9207	0.8402	0.8604	0.8384	2.72	1.74	0.1065
RF	n=300, d=20	0.8985	0.918	0.836	0.8763	0.8429	2.73	1.81	0.1737
GBDT	n=200,d=6,η=0.05	0.8973	0.917	0.8351	0.8589	0.8446	2.75	1.84	0.1032
ExtraTrees	n=300,d=20	0.8954	0.9142	0.8326	0.871	0.844	2.77	1.83	0.1384
CatBoost	n=300,d=6,η=0.05	0.8578	0.8801	0.772	0.7899	0.8125	3.24	2.25	0.0962
MLP	[64,32],ReLU	0.7619	0.7833	0.6031	0.7009	0.757	4.19	2.85	0.0204
SVM	RBF,C=1.0	0.6574	0.6729	0.5409	0.4827	0.6885	5.02	3.35	0.6483
Ridge	α=1.0	0.6097	0.6369	0.4279	0.3984	0.6278	5.36	3.79	-0.0406

To screen for the optimal predictive model capable of precisely mapping the complex

non-linear relationships between the "environment-substrate" coupled features and the agronomic trait development indicators of *Lentinula edodes*, this study systematically compared 10 algorithms, including Ridge, KNN, SVM, RF, ExtraTrees, GBDT, XGBoost, LightGBM, CatBoost, and MLP. As shown in Table 2, both the LightGBM and XGBoost models achieved good performance in the prediction tasks of the four agronomic trait targets, but XGBoost performed better in terms of MBE. Therefore, considering specific conditions, XGBoost was selected as the model for subsequent SHAP explanations.

Prediction Performance of XGBoost Model with 95% Confidence Intervals

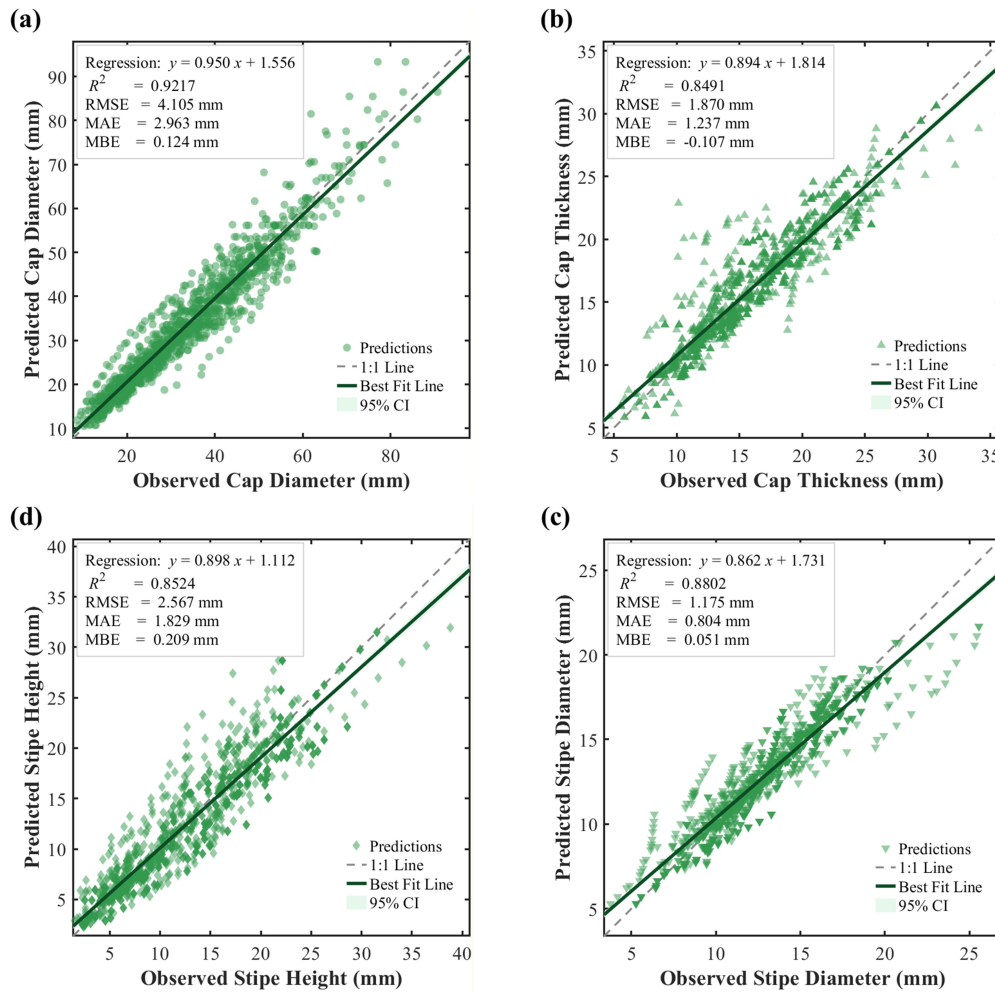


Fig. 5 Prediction performance and 95% confidence intervals of the XGBoost model

Compared with other models, XGBoost demonstrated outstanding generalization ability and robustness. Its R^2 for the four parameter indicators of the mushroom reached as high as 0.904, 0.92, 0.85, and 0.88, respectively. This is primarily attributed to XGBoost's innovative introduction of a second-order Taylor expansion and a regularization penalty term in its objective function, which can effectively control the complexity of the tree model, thereby effectively suppressing overfitting when processing agricultural sensory data with multicollinearity (Chen & Guestrin, 2016). Reliable model explainability must be established on the foundation of high predictive accuracy. Given that XGBoost successfully captured the genuine physical mapping laws between the input features and the agronomic trait develop

ment of the mushroom, this study ultimately selected it as the core predictor for the subsequent attribution analysis.

To break the unexplainability of high-dimensional black-box models, this study introduced the SHAP (SHapley Additive exPlanations) framework based on game theory (Lundberg & Lee, 2017), and adopted the TreeExplainer algorithm specifically optimized for ensemble tree models (Lundberg et al., 2020). This algorithm can accurately isolate and quantify the global importance and local interactive effects of environmental microclimates and substrate physicochemical factors on various agronomic trait dimensions of the mushroom, thereby providing data-driven explanations with physical mechanistic significance for the precise environmental control of mushrooms.

3.2 SHAP Result

3.2.1 Distinct Target Regulatory Paths of Substrate Nutrition

Global feature attribution via the SHAP framework deciphers the underlying multi-dimensional driving forces governing *Lentinula edodes* agronomic trait (Fig. 6). The global importance ranking establishes a clear divergence in how the 11 screened physicochemical and environmental variables partition their regulatory control across different agronomic trait anatomical regions.

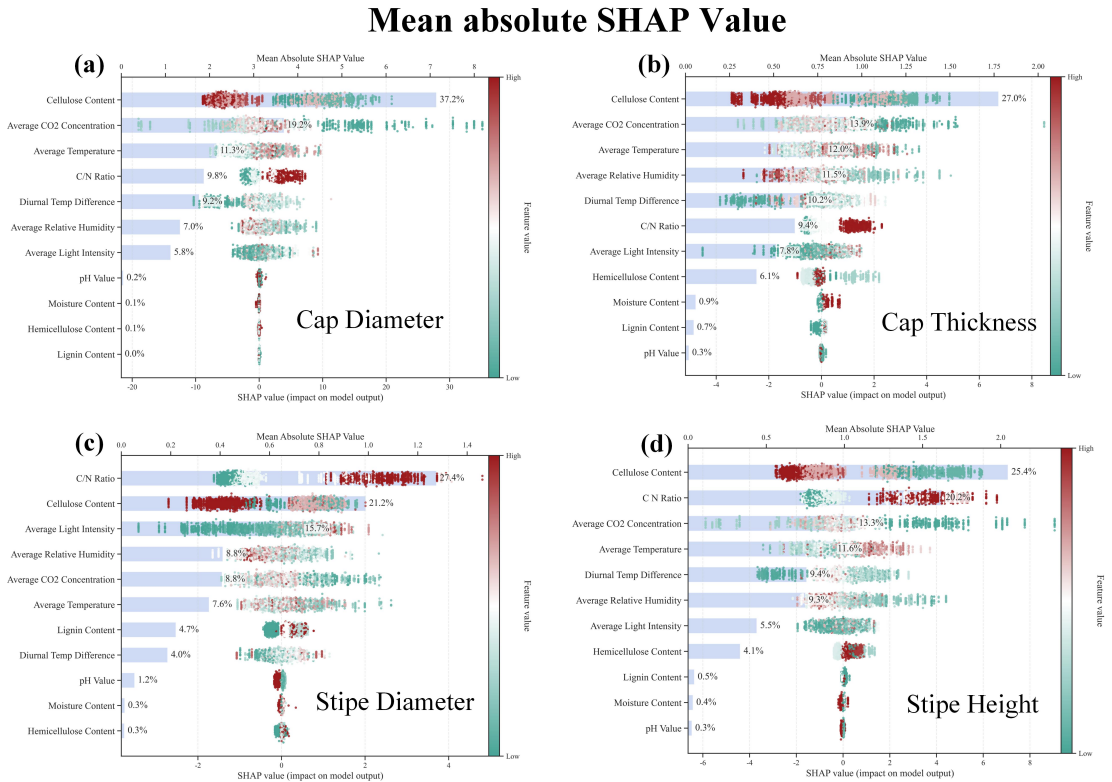


Fig. 6 SHAP beeswarm plots for *Lentinula edodes* agronomic trait prediction models

The substrate physicochemical matrix serves as the fundamental template for phenotypic determination, exhibiting strict tissue targeting. (Gaitán-Hernández et al., 2011). Cellulose content acts as the primary restrictive driver for lateral pileus expansion, commanding the highest feature importance at 37.2% for cap diameter and 27.0% for cap thickness. High local concentrations of cellulose (represented by red ensembles) systematically shift the baseline toward the negative

domain of SHAP values, demonstrating a pronounced physiological constraint on both the outward expansion and thickening of the mushroom cap meat. (Baktemur et al., 2022).

Concurrently, microclimatic dynamics modulate the physiological baseline via non-linear threshold controls. Average CO_2 concentration and ambient temperature maintain persistent feature attributions (ranging from 11% to 19%) across all four targets. Rather than manifesting monotonic linear dependencies, the wide, non-uniform spatial scattering of these environmental attributions reinforces that microclimatic regulation is governed by intricate threshold-driven networks and non-linear multi-variable coupling. (Jang et al., 2013).

3.2.2 Comparison of Main and Interaction Effects of Features

To further dissect the influence mechanisms of each feature variable on agronomic trait, the contribution of each feature to the prediction outcomes was decomposed into independent main effects and synergistic interaction effects (Fig.7). By comparing these two metrics, the operational modes of each variable during mushroom development can be clearly demarcated (Lundberg & Lee, 2017):

All Features : Main vs Interaction

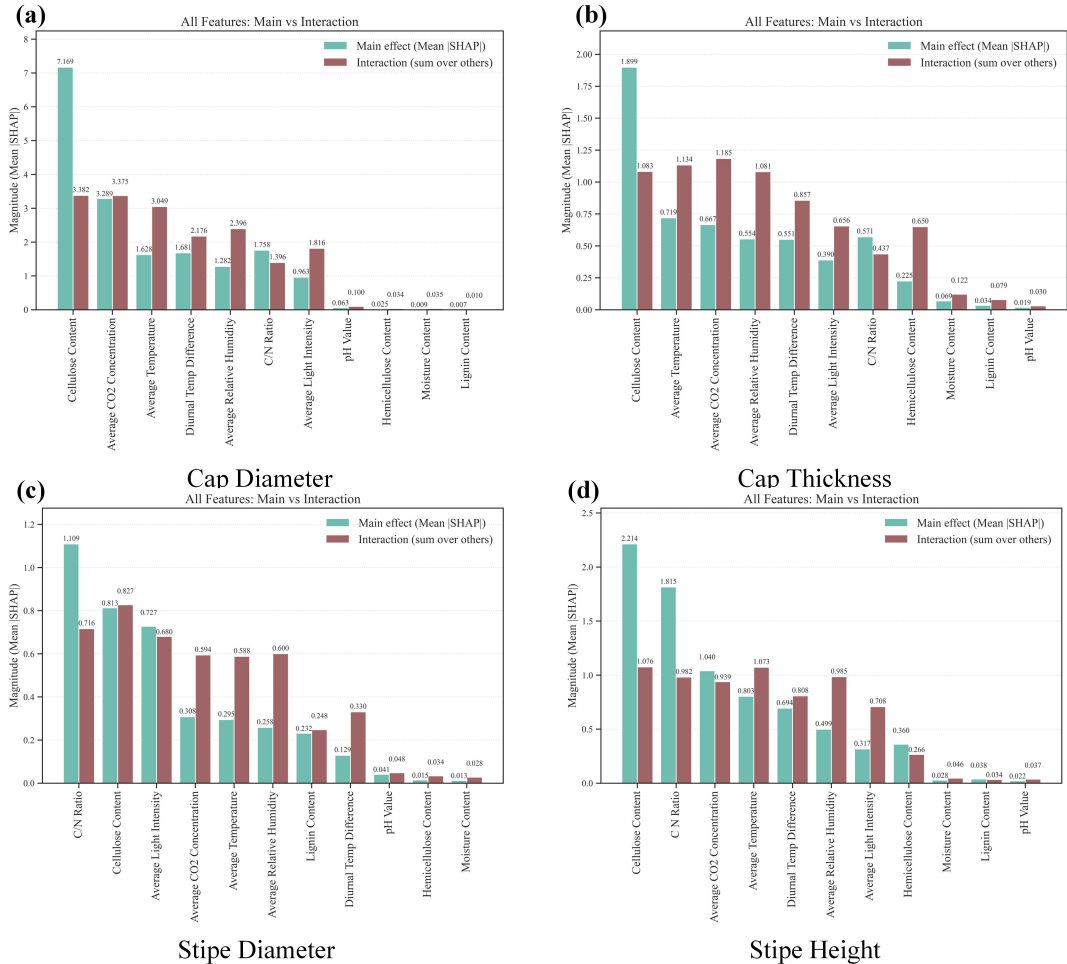


Fig. 7 SHAP main and interaction effects of features on *Lentinula edodes* agronomic trait

Decomposing the global SHAP attributions into independent main effects and high-order synergistic interactions reveals the explicit operational modalities of distinct features (Fig. 7). (Lundberg & Lee, 2017) Substrate nutrition conforms to an "independently dominant and d

irectly driven" paradigm. For cellulose content and the substrate *C/N* ratio, the isolated main effects significantly outpace their corresponding interaction terms across key target variables. Specifically, the main effect of cellulose on cap diameter reaches a peak magnitude of 7.169, verifying a highly robust, direct metabolic pathway that remains resilient against external microclimatic perturbations. (Gaitán-Hernández et al., 2011).

In stark contrast, environmental modulators operate through a "highly coupled and synergistically regulated" mechanism. For microclimatic variables—including average temperature, CO_2 concentration, and relative humidity—the interaction effects frequently equal or surpass their independent main effects. This structural "inversion" is particularly pronounced in the cap thickness and stipe diameter networks, where the collective interaction magnitude exceeds the main effects, highlighting that single microclimatic elements rarely dictate morphogenesis in isolation but function as systemic co-catalysts. (Subashini et al., 2024).

3.2.3 Non-linear Synergistic Networks Between Carbon/Nitrogen Sources and Microclimates

To deeply explore the specific driving mechanisms behind different agronomic trait indicators, Figure 7 details the SHAP interaction effect matrices for the four core agronomic trait features of the cap and stipe. By horizontally comparing the interaction extreme values (dark red grids) outside the diagonal, it can be found that the morphogenesis of different parts of the mushroom is controlled by different variable coupling networks:

Cap development: Relies on strong synergy between "carbon source-stomata" and "microclimate" Cap diameter (Fig.8a): The matrix shows that the interaction strength between cellulose content and average CO_2 concentration reached an extremely high 1.026, far exceeding other feature pairs. This indicates that the conversion efficiency of high carbon sources in the substrate highly depends on specific environmental carbon dioxide concentrations, and the strong coupling of the two is the core driving force promoting the lateral expansion of the cap (Gaitán-Hernández et al., 2011). Cap thickness (Fig.8b): Unlike diameter, the development of thickness relies more on the synergy of the physical microclimate. Average relative humidity and average temperature (0.274), as well as hemicellulose and average CO_2 concentration (0.242), exhibit the strongest interaction effects, indicating that temperature and humidity synergistically dominate the vertical thickening and dry matter accumulation process of the cap flesh (Jang et al., 2013).

Stipe development: Centers on nutrient and environmental interaction around the "*C/N* ratio" Stipe diameter (Fig.8c): The internal synergy of substrate nutrition is dominant. Cellulose content and the *C/N* ratio (0.199) constitute the most significant interaction pair; combined with the distribution pattern of the scatter plots, it reflects that a reasonable ratio of basic substrate nutrition is a prerequisite for determining the stoutness of the stipe (Baktemur et al., 2022). Stipe height (Fig.8d): Stipe elongation exhibits typical "substrate-based, environment-catalyzed" characteristics. Average CO_2 concentration and the *C/N* ratio (0.239), and average temperature and cellulose (0.215) show strong coupling. This means that under specific substrate nutritional bases, the dynamic intervention of environmental factors (especially temperature control and ventilation) is the key trigger switch inducing the vertical elongation of the stipe (Subashini et al., 2024).

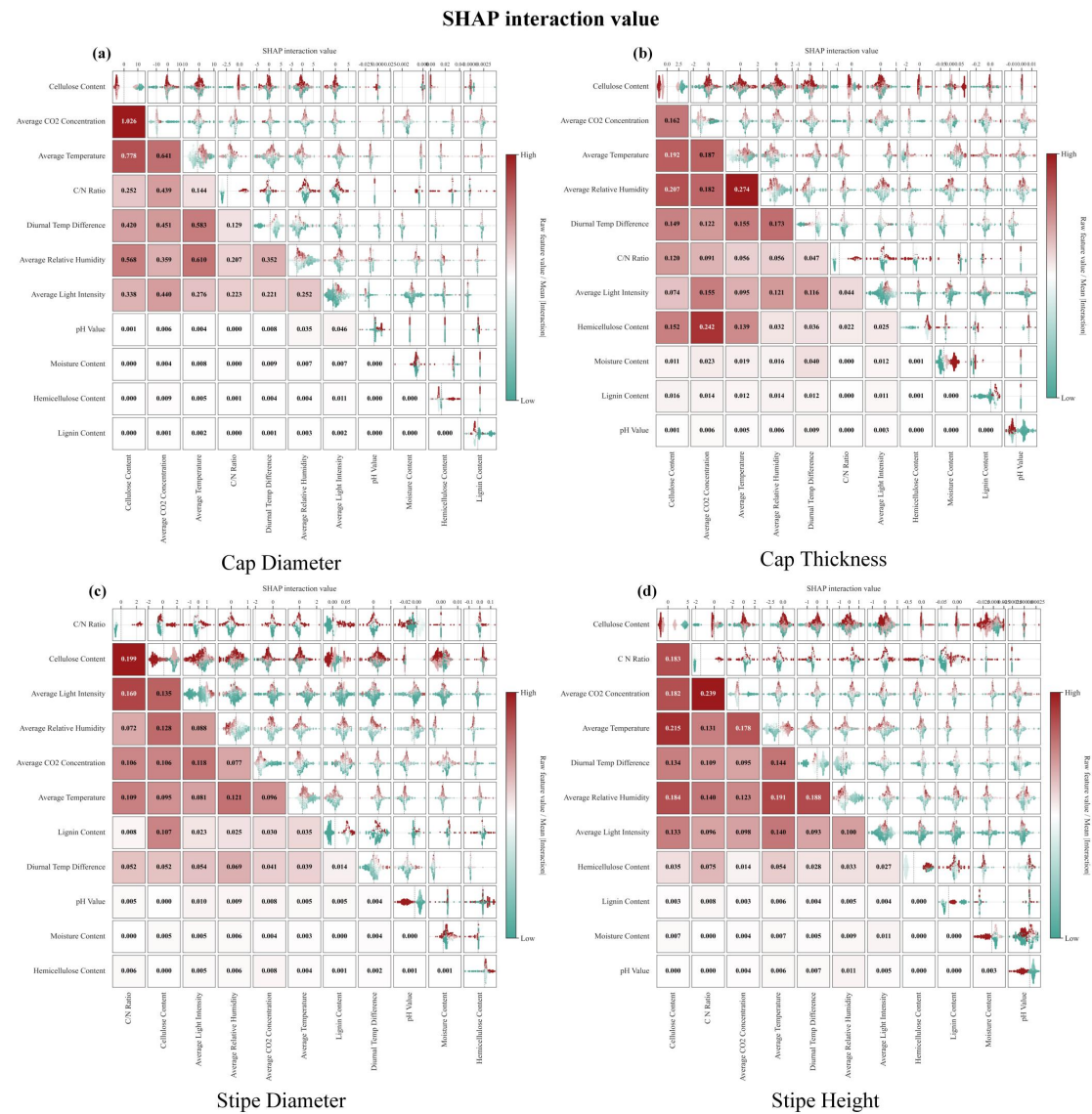


Fig. 8 Global SHAP interaction heatmap and distribution among environmental and substrate parameters

3.2.4 Topological Evolution of "Environment-Substrate" Regulatory Hubs

To globally characterize the systematic topological structure of the "environment-substrate" variables during mushroom development, this paper introduces the SHAP feature impact and interaction network graph (Fig. 9) (Lundberg et al., 2020). This network graph abstracts variables as nodes (node size reflects the main effect strength) and the synergistic relationships between variables as connecting lines (line thickness represents interaction strength), visually revealing the systematic driving modes behind different agronomic trait features:

We can visually observe that the cap development network is a pattern shifting from "single-core radiation" to "multi-point synergy". Cap diameter (Fig.9a): Exhibits a typical "single-core radiating" topological structure. Cellulose content is the absolute mega-hub node in the network; not only does its own main effect occupy absolute dominance, but it also radiates multiple strong connections (thick red lines) outward, especially with average CO₂ concentration and average temperature. This indicates that the lateral expansion of the cap is a process fundamentally driven by a single carbon source and directionally amplified by the

microenvironment. Cap thickness (Fig.9b): The network evolves into a "multi-center coupling" structure. The absolute advantage of the cellulose node is weakened, while the nodes of environmental factors such as relative humidity, average temperature, and diurnal temperature difference are significantly enlarged. A dense network of strong connections is formed among these environmental factors and with cellulose, indicating that cap thickening no longer relies on a single hub, but is the systematic result of deep synergy between the environmental microclimate and the substrate (Jang et al., 2013).

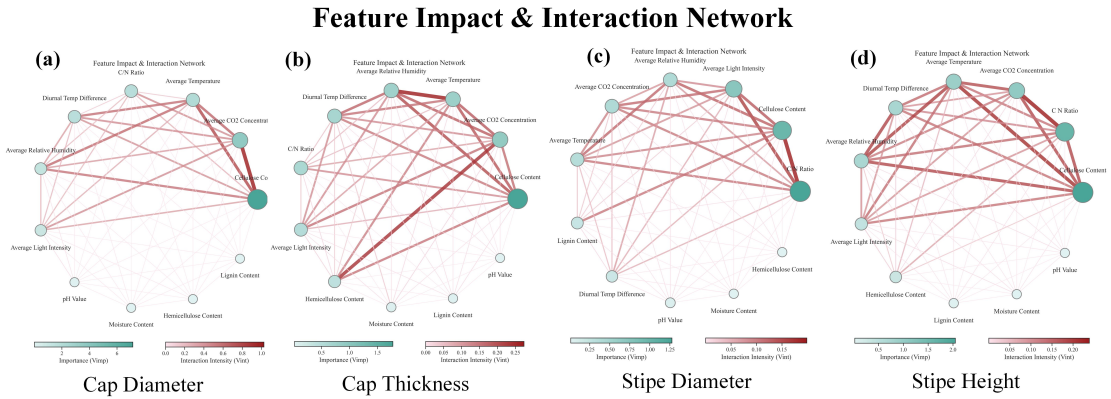


Fig. 9 SHAP feature impact and interaction networks

For the stipe development network, it represents nutrient remodeling and a "dual-core full-network" interaction. Stipe diameter (Fig.9c): The network core has shifted significantly, with the carbon-to-nitrogen ratio (C/N Ratio) replacing cellulose as the largest hub node in the system. The most prominent structural feature in the graph is the extremely thick connecting line between the C/N ratio and cellulose, which visually proves that the "carbon-nitrogen nutrient balance" and the internal synergy of carbon source degradation within the substrate constitute the most core mechanism determining stipe thickening (Baktemur et al., 2022). Stipe height (Fig.9d): Presents a highly complex "dual-core, full-network interconnected" topological feature. Cellulose and the C/N ratio, acting as dual core nodes of the substrate, establish high-strength reticular connections with almost all environmental nodes such as temperature, humidity, and CO₂. This reveals the extreme sensitivity of stipe elongation to environmental fluctuations, representing a complex process involving the comprehensive coupling of the substrate nutritional base and the microclimate.

3.2.5 Individual Sample Contribution Analysis

To deeply penetrate the microscopic decision-making process of the predictive model in the morphogenesis of different individual mushrooms, Fig. 10 illustrates the individual sample attribution heatmap based on SHAP values. This plot performs a horizontal topological mapping of the local explanations of global samples: the top curve visually displays the fluctuation trajectory of each sample's predicted value relative to the model baseline, while the alternating warm and cool color bands below meticulously characterize the dynamic game process of specific features pushing up (red positive area) or pulling down (green negative area) the final predicted value of the individual.

SHAP instance heatmap of predictive features

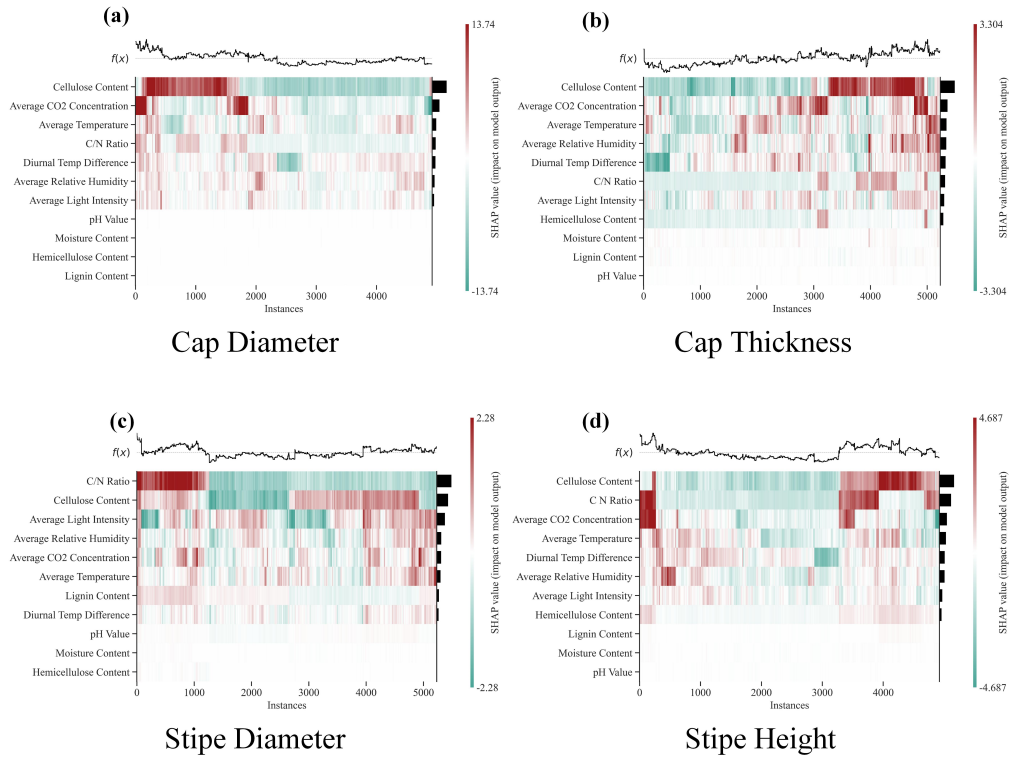


Fig. 10 SHAP instance heatmap of predictive features

Through a deep analysis of the sample spatial distribution across the four agronomic trait indicators, the following individual-level driving patterns can be derived:

1. Local Driving Differences in Cap agronomic trait Prediction

Cap diameter (Fig.10a): The predictive network exhibits significant polarization characteristics. Among the sample intervals with high predicted values (left side of the plot), cellulose content and average concentration burst with extremely strong positive driving forces (highly concentrated dark red blocks). However, as drops stepwise, the contributions of these two core features rapidly undergo a threshold flip, transforming into strong negative inhibition (green). This directly explains the microscopic mechanism by which specific individuals grow large-sized caps: it is highly dependent on the highly efficient combined pull of carbon source conversion and carbon metabolism in the localized microenvironment.

Cap thickness (Fig.10b): Developmental heterogeneity at the individual level is significantly enhanced. In the sample group driving thickness growth (rising curve), the absolute dominant position of a single feature is weakened. Instead, cellulose content, average temperature, and relative humidity alternately present red positive thrusts across different sample intervals. This indicates that the microscopic triggers for individual cap thickening possess diversity, representing a complex systematic process regulated sequentially by multiple microclimatic factors adapting to local conditions.

2. Core Thrust Locking in Stipe agronomic trait Prediction

Stipe diameter (Fig.10c): The predictive logic of the model demonstrates a high degree of focus at the individual level. In the high prediction value intervals promoting stipe thickening, the C/N ratio releases an overwhelming positive thrust. When the C/N ratio of

a specific sample deviates from the optimal range, this feature immediately manifests a strong inhibitory effect, directly suppressing the prediction curve below the baseline. This fully proves that the core physical "criterion" determining the stoutness of an individual stipe is highly locked onto the nutritional balance state of the substrate(Baktemur et al., 2022).

Stipe height (Fig.10d): Accurately reveals the developmental mechanism of extreme deformed morphologies. Among the extreme sample groups causing sudden surges in stipe height (right-side peak area of the plot), cellulose content and the C/N ratio synchronously present high-intensity dark red positive contributions, accompanied by the synergistic push of microenvironmental concentration. This finding precisely pinpoints the individual microscopic triggers of the "spindly stipe" defect in production: namely, the imbalance of the dual-core nutritional indicators of the substrate, superimposed with the catalysis of specific microclimatic gases, jointly push the individual's vertical elongation limit to the extreme.(Moon et al., 2024)

4. Discussion

4.1 Advantages of Multi-dimensional Spatiotemporal Modeling for Greenhouse IoT Big Data

The paramount challenge in modern factory mushroom cultivation lies in the spatial heterogeneity and non-linear dynamics of the microclimate (Bojacá et al., 2009). This study demonstrates that among the 10 machine learning frameworks compared, XGBoost exhibited the optimal anti-noise robustness. This is primarily attributed to its built-in second-order Taylor expansion and regularization penalty terms, which effectively filter high-frequency noise from underlying IoT sensor networks (Chen & Guestrin, 2016). Particularly when processing continuous sensory data spanning from winter to early spring (December to March)—a period characterized by severe thermodynamic fluctuations and extreme spatial gradients—traditional global mean models are highly susceptible to failure. Conversely, XGBoost, coupled with IDW spatial interpolation reconstruction techniques, successfully eliminated systematic prediction biases. This high-fidelity digital modeling approach overcomes the physical limitations of discrete sensors, offering a highly promising underlying benchmark engine for constructing factory "digital twin" systems for edible fungi (Pyliaidis et al., 2021).

4.2 Microscopic Targeted Attribution of Carbon-Nitrogen Imbalance and the "Spindly Stipe" Defect

Historically, the regulation of morphogenesis in shiitake cultivation has heavily relied on macroscopic empirical knowledge, lacking quantitative support at the mechanistic level. Leveraging the SHAP framework (Lundberg & Lee, 2017), this study defined the core control targets at a microscopic quantitative level for the first time: cellulose content dominates the lateral expansion of the cap, while the carbon-to-nitrogen ratio (C/N) dictates the vertical development of the stipe. This quantitative mapping aligns perfectly with the carbon-nitrogen metabolism diversion theory in fungal physiology (Baktemur et al., 2022).

More importantly, this study precisely pinpointed the microscopic triggers for the common "spindly stipe" defect in production. The SHAP interaction network analysis revealed that this

extreme agronomic trait defect is not caused by singular environmental deterioration, but is rather a vicious coupled outcome triggered by dual-core substrate nutritional indicators (i.e., C/N imbalance and excessive cellulose consumption) superimposed with specific greenhouse thermodynamic oscillations and ventilation anomalies (localized CO₂ retention) (Moon et al., 2024). This multidimensional attribution provides a clear theoretical basis for bidirectionally preventing agronomic trait defects from both the source of substrate formulation and the terminal of environmental control.

4.3 A New Paradigm of Targeted Environmental Intervention and Industrial Implications

At the agronomic mechanism level, this study confirms that modern factory mushroom cultivation must completely abandon the traditional "single macro-environmental mean management" mode (Ghazal et al., 2024). Analysis indicates that microclimatic factors primarily act as sensitive "catalysts" and synergistic amplifiers in agronomic trait development rather than singular decisive factors. Therefore, future practical production should transition toward a targeted intervention paradigm of "customizing strategies based on goals" (Sharma et al., 2020).

Specifically, if the cultivation goal is to produce high-grade, large-sized caps, emphasis should be placed on coordinating the spatiotemporal matching between the underlying cellulose degradation rate and stomatal conductance (regulated by microenvironmental CO₂ concentration). Conversely, if the aim is to suppress spindly stipes and cultivate stout mushrooms, the dynamic coupling between the substrate carbon-nitrogen base and the spatial 3D temperature and humidity fields must be strictly monitored. This data-driven, agronomic trait-directed regulatory framework establishes a crucial scientific decision-making pathway for the intelligent and high-value transformation of the edible fungi industry.

5. Conclusion

Addressing the persistent pain points of unclear multi-dimensional environmental coupling mechanisms and the lack of directional regulation bases in the factory cultivation of *Lentinula edodes*, this study proposed and validated a morphology-directed attribution framework integrating IoT spatial reconstruction, machine learning multi-model ensembles, and Explainable AI (XAI). The main conclusions are as follows:

1. **Established a high-fidelity agronomic trait prediction benchmark:** Among 10 mainstream machine learning architectures, the XGBoost algorithm performed optimally when processing agricultural sensory data with severe spatiotemporal heterogeneity. Its predictive coefficients of determination R^2 for the four phenotypes—cap diameter, cap thickness, stipe diameter, and stipe height—reached 0.9217, 0.8491, 0.8802, and 0.8524, respectively, qualifying it as a reliable predictive core for edible fungi digital twin systems.(Chen & Guestrin, 2016)
2. **Quantified the core control targets of phenotypic development:** Substrate physicochemical parameters are the fundamental dominant factors determining agronomic trait and exhibit strict targeting. Cellulose content is the core source driving lateral cap expansion (feature contribution rate 37.2%), while the substrate C/N ratio is the absolute hub controlling vertical stipe development (contribu

tion rate 27.4%), providing quantitative standards for substrate formulation optimization(Baktemur et al., 2022)..

3. **Revealed the trigger mechanisms of extreme agronomic trait defects:** Microclimatic factors act as sensitive synergistic catalysts within the developmental network. The individual microscopic triggers for the "spindly stipe" excessive growth defect were precisely defined as the strong coupled superimposition of substrate C/N imbalance with localized greenhouse thermodynamic and CO₂ oscillations.
4. **Innovated a data-driven cultivation management paradigm:** This study proved the necessity of abandoning macroscopic mean management and established a targeted 3D spatial regulation mode of "customizing strategies based on goals." This model provides a scientific decision-making tool with immense engineering application value for the edible fungi industry's transformation from "empirical blind planting" to "data-driven targeted phenotypic cultivation."(Ghazal et al., 2024).

CRedit authorship contribution statement

Liu Weihao:Writing-original draft, Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization. **Liu Tianyi:**Formal analysis, Investigation, Software, Validation. **Xiao Jinyu:**Data curation, Investigation. **Bi Ziyao:** Formal analysis, Visualization. **Gong Yuhua:**Investigation, Resources. **Tan Zuojun:**Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing-review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A.Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.57760/sciencedb.38980>

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